**RESUME**

# JingJiang


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13041407150

1603165697@qq.com

JiangSu, SuZhou


## EDUCATION

2024.03 – Present Institute of Systems Medicine, Suzhou Bioinformatics Technician

Job Responsibilities:

- Responsible for all bioinformatics analysis and statistical computations within the group, mainly focusing on the tRNA field. Utilizing second-generation sequencing and third-generation sequencing analysis, as well as single-cell transcriptome analysis techniques to investigate the impact of viruses on the organism.

2021.09-2024.06 Yangzhou University Master of Experimental Animals and Comparative Medicine

Graduation Design:

- The Functional and Molecular Mechanisms of Endogenous Retroviruses in Placental Transcriptional Regulatory Networks in Humans and Mice

Key Points of the Design:

- Utilized data from public databases such as NCBI, Ensembl, and UCSC for bioinformatics analysis, confirming the placenta-specificity of lineage-specific enhancers across different species, providing evidence for transcriptional regulation specificity associated with ERV elements.

2017.09-2021.06 Yangzhou University Bachelor of Animal and Plant Inspection

## RESEARCH RESULTS

Currently Published Papers:

- Du C, **Jiang J**. Regulation of endogenous retrovirus-derived regulatory elements by GATA2/3 and MSX2 in human trophoblast stem cells. Genome Res.2023(co-first author)
- Weiyun Qin; **Jing Jiang**; Yunxi Xie; Zhengchang Wu; Ming-an Sun; Wenbin Bao. Exosomal ssc-miR-1343 targets FAM131C to regulate porcine epidemic diarrhea virus infection in pigs. Veterinary Research.2024 (co-first author)
- **Jiang J** et al. Endogenous retrovirus-derived enhancers confer the transcriptional regulation of human trophoblast syncytialization. Nucleic Acids Res. (no co-first author)
- Zhang L, Jin J, Qin W, **Jiang J** , Bao W , Sun M. Inhibition of EZH2 causes retrotransposon derepression and immune activation in porcine lung alveolar macrophages. Int J Mol Sci.

## SPECIAL SKILLS

- Analysis Skills:

  Proficient in pathway analysis and transcriptome analysis upstream and downstream analysis (using linux, R, NGS analysis)

  Transcription factor analysis and visualization (igv, TEtranscript, deepTools)

  RNA editing, DNA methylation analysis, circRNA/lncRNA/trRNA related analysis (cirIquant, stringtie)

- Experiment Skills:

  Familiar with hTSC, mTSC cell culture/differentiation, extraction of mouse primary cells, chip-seq experiment, qPCR, WB and other basic experiments.

## PARTICIPATED IN THE PROJECT

- Participated in the project "Function and Regulatory Mechanism of MER41 Transposon in Human Placental Transcriptional Regulatory Networks" (National Natural Science Foundation of China General Program)
- Participated in the Jiangsu Province "Distinguished Professor" program and the Jiangsu Province "Innovative and Entrepreneurial Team" program
- Utilized nanopore sequencing technology to study mechanisms related to tRNA.
- Other Participated Projects:
  - ➢ Third-generation sequencing (Nanopore) result analysis

  Content: In addition to routine analysis, involved the application of various software tools (nanorms, nanospa, tai, rscu, etc.) and statistical analysis of the results.
  - ➢ Single-cell transcriptome result analysis

  Content: Employed AI techniques to integrate bulk RNA-seq data with single-cell data, converting bulk RNA-seq data into semi-single-cell sequencing data.
  - ➢ Combined second-generation and third-generation sequencing result analysis

  Content: Integrated article analysis to compare pre-tRNA and mature-tRNA alignment results.
  - ➢ Combined riboseq and RNAseq data result analysis

  Content: Conducted codon pause analysis while removing the influence of mRNA expression levels to examine the impact of tRNA on codon pause analysis.
  - ➢ Clipper data analysis (Completed specific requirement analyses for different research groups).
  - ➢ Amplicon data analysis